Exp 1: Design and Optimization of Half-Wave Dipole Antenna using MMANA-GAL

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Aim: To design a half-wave dipole at 300 MHz and optimize its length for minimum VSWR and maximum gain.

Design Parameters

Frequency	Wavelength	Dipole length	Wire radius	Segments	Source
300 MHz	1 m	0.478 m (half-length 0.239 m)	1 mm	22	w1c, 10 V

Geometry Table (MMANA-GAL)

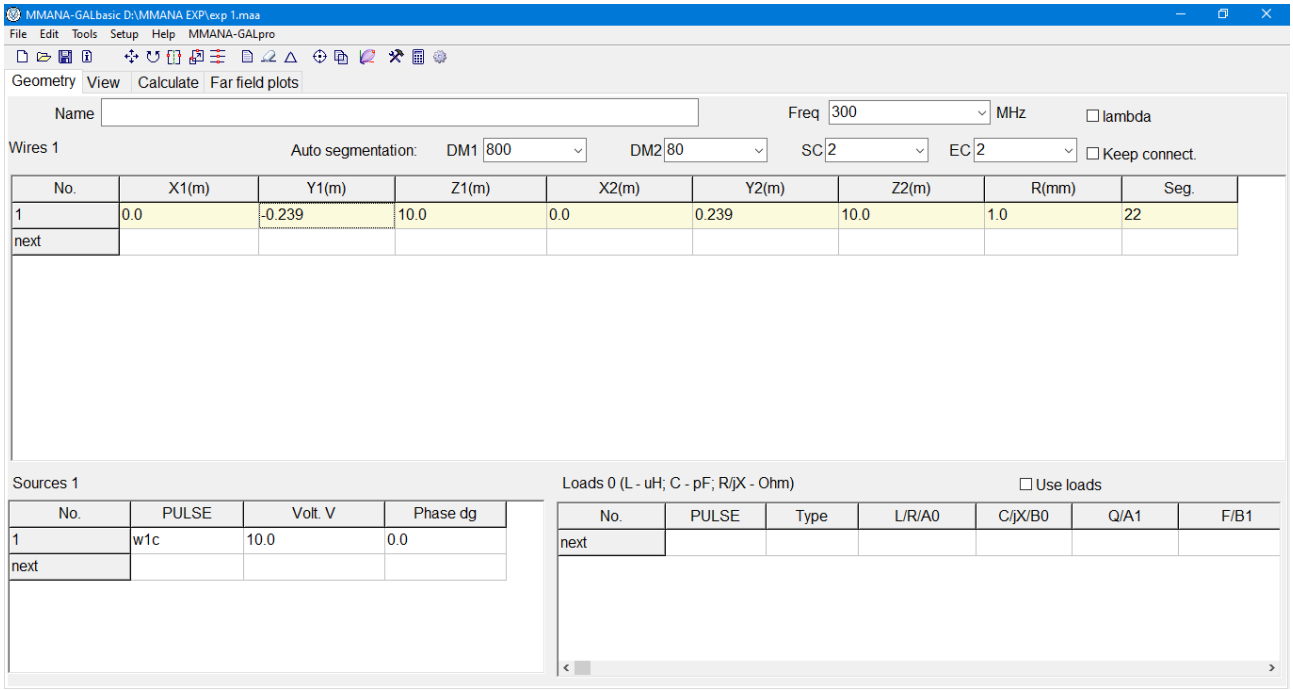


Fig. 1: Geometry tab of the dipole (wire along Y, feed at the centre of wire 1)

Far Field Radiation Pattern

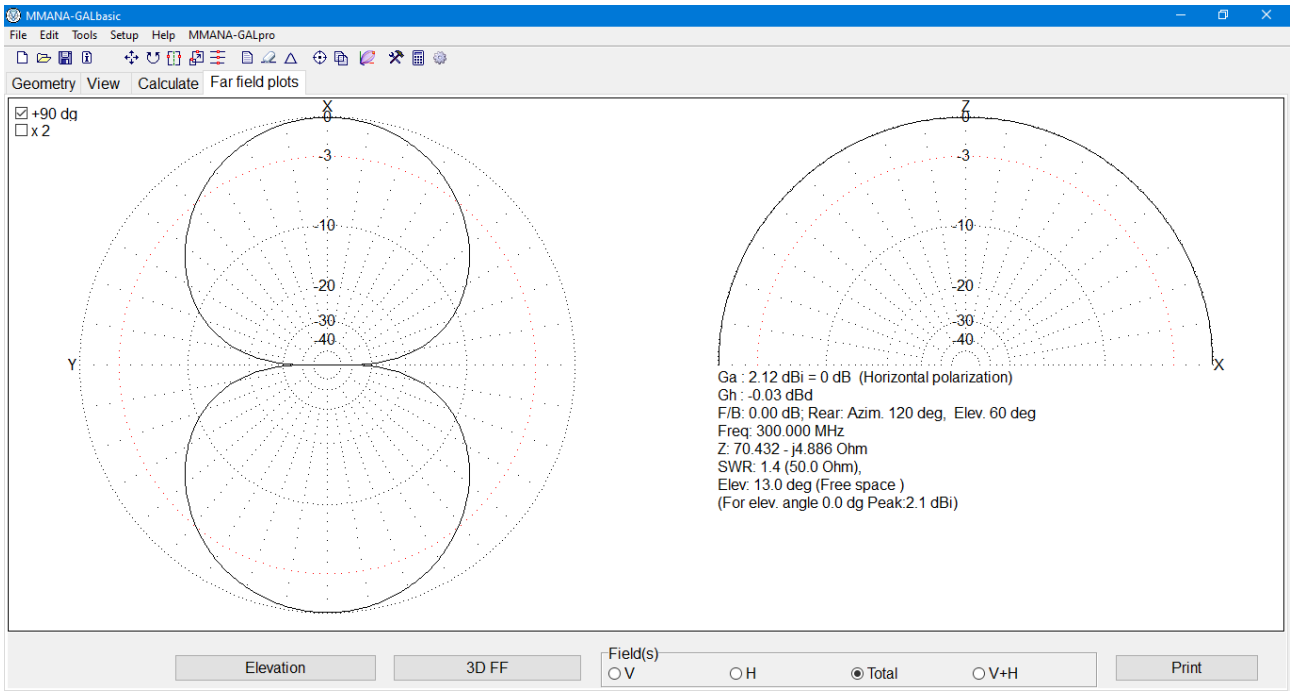


Fig. 2: Far field plot of the dipole. Left: azimuth (figure-8 pattern). Right: elevation.

Results (from the far field plot)

Gain (dBi)	Impedance (Ohm)	SWR (50 Ohm)	F/B (dB)
2.12	70.43 - j4.886	1.4	0.00